

SPRINGFIELD-BRANSON, MO, USA

WMO: 724400

Lat: 37.2347N

Lon: 93.4014W

Elev: 390

StdP: 96.73

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.8	-19.1	0.7	-12.2	-16.3	0.9	-9.2	11.9	5.9	11.0	5.1	3.6	340	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	35.2	23.4	33.5	23.5	32.1	23.4	25.4	31.6	24.9	30.9	24.2	30.0	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.4	28.6	23.1	18.7	28.1	22.5	18.0	27.4	80.1	31.6	77.5	30.8	75.0	30.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.2	8.4	-18.1	36.9	3.2	1.8	-20.4	38.3	-22.3	39.3	-24.1	40.4	-26.4	41.7
			-18.6	26.3	3.1	0.8	-20.8	26.9	-22.6	27.3	-24.3	27.7	-26.5	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	1.0	3.4	8.3	13.7	18.6	23.6	25.8	25.4	21.1	14.6	8.2	2.6
	DBStd	10.03	6.61	6.37	6.05	5.01	4.36	3.14	2.76	2.99	4.20	5.03	5.42	5.82
	HDD10.0	963	288	199	103	22	0	0	0	0	16	97	237	
	HDD18.3	2427	539	418	312	152	52	3	0	1	21	137	306	487
	CDD10.0	2392	7	14	52	133	269	407	491	477	332	158	41	9
	CDD18.3	814	0	0	2	13	61	161	233	219	103	21	1	0
	CDH23.3	7602	0	2	21	106	452	1461	2374	2142	882	153	8	0
	CDH26.7	3073	0	0	1	13	103	554	1074	978	319	31	0	0
(14) Wind	WSAvg	4.1	4.5	4.6	4.9	4.8	4.1	3.8	3.2	3.2	3.5	4.0	4.4	4.4
(15) Precipitation	PrecAvg	1103	52	58	92	112	120	122	87	91	113	88	91	76
	PrecMax	1576	173	192	239	306	341	341	234	218	444	255	311	291
	PrecMin	695	2	2	22	4	39	6	8	9	3	11	5	11
	PrecStd	213	36	41	50	60	57	68	54	48	83	55	67	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	22.6	25.6	28.4	30.8	34.5	37.2	37.3	34.5	29.8	24.3	19.9
		MCWB	13.5	14.3	16.7	18.5	22.1	23.6	22.7	22.9	22.3	21.1	17.0	15.0
	2%	DB	16.9	18.9	23.1	26.3	29.2	32.7	35.2	35.4	32.1	27.1	21.2	16.9
		MCWB	11.8	12.5	15.2	17.4	21.2	23.4	23.6	23.4	22.3	19.1	15.2	13.0
	5%	DB	14.0	16.4	20.4	24.2	27.7	31.3	33.4	33.6	30.3	24.9	19.0	14.1
		MCWB	9.9	11.3	13.7	16.6	20.7	23.1	23.9	23.4	21.8	17.9	14.3	10.5
	10%	DB	10.8	13.7	18.1	22.3	26.1	30.0	32.0	31.8	28.3	22.7	16.8	11.5
		MCWB	7.4	9.4	12.4	15.6	19.9	22.7	23.9	23.2	20.6	17.2	12.7	8.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	16.5	18.2	20.2	23.6	25.6	26.3	25.9	24.4	22.3	18.6	16.4
		MCDB	18.5	20.6	23.8	25.7	29.0	31.7	32.2	32.6	30.4	27.4	22.4	19.2
	2%	WB	13.2	14.4	16.4	19.1	22.6	24.8	25.6	25.2	23.5	20.8	16.7	13.8
		MCDB	15.7	17.7	21.8	24.0	27.6	30.8	31.6	31.7	29.6	24.6	19.8	16.2
	5%	WB	10.4	12.0	14.9	17.9	21.5	24.0	25.1	24.6	22.7	19.6	15.1	11.2
		MCDB	13.3	15.1	19.2	22.3	26.3	29.6	31.0	30.8	28.5	23.1	18.0	13.2
	10%	WB	7.4	9.4	13.2	16.7	20.5	23.3	24.5	23.8	21.9	18.1	13.3	8.8
		MCDB	11.0	13.6	17.2	21.0	24.9	28.5	30.1	29.8	26.7	21.6	16.3	11.0
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	11.6	11.7	10.6	10.5	10.6	11.0	11.4	11.5	11.0	9.8
		MCDBR	14.5	14.7	14.5	13.7	11.6	11.7	12.4	13.2	12.5	12.9	13.4	13.2
	5% WB	MCWBR	9.9	9.8	8.6	7.6	5.8	4.8	4.1	4.2	4.9	6.5	8.8	9.5
		MCWBR	12.5	13.0	12.2	11.4	10.5	10.6	10.6	11.0	10.8	10.6	11.7	11.6
(40) Clear-Sky Solar Irradiance	taub	0.299	0.307	0.339	0.376	0.414	0.422	0.440	0.438	0.380	0.340	0.307	0.291	
		taud	2.467	2.451	2.406	2.321	2.272	2.278	2.275	2.257	2.392	2.483	2.520	2.513
	Ebn at Noon	894	930	926	904	871	860	841	836	870	880	878	881	
		Edh at Noon	82	94	109	126	134	134	133	110	91	78	74	
(44) All-Sky Solar Radiation	RadAvg	2.37	2.96	3.96	5.03	5.64	6.46	6.32	5.71	4.89	3.58	2.52	1.99	
	RadStd	0.23	0.39	0.24	0.29	0.45	0.51	0.40	0.37	0.37	0.41	0.27	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page